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Association between indoor formaldehyde exposure and asthma: A systematic review and meta-analysis of observational studies

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Abstract

About 339 million people worldwide are suffering from asthma. We aimed to investigate whether exposure to formaldehyde (FA) is associated with asthma, which could provide clues for preventive and mitigation actions. This article provides a systematic review and meta-analysis of observational studies to assess the association between indoor FA exposure and the risk of asthma in children and adults. An electronic search of PubMed, Embase, and Web of Science was performed to collect all relevant studies published before January 1, 2020, and a total of 13 papers were included in this meta-analysis. A random-effect model was conducted to calculate the pooled odds ratio (OR) between FA exposure and asthma. We found that each 10 $\mu\text{g}/\text{m}^3$ increase in FA exposure was significantly associated with a 10% increase in the risk of asthma in children (OR = 1.10, 95% confidence interval = 1.00–1.21). We sorted the FA concentrations reported in the selected articles and categorized exposure variables into low (FA $\leq 22.5 \mu\text{g}/\text{m}^3$) and high exposure (FA $> 22.5 \mu\text{g}/\text{m}^3$) according to the median concentration of FA. In the high-exposure adult group, FA exposure may also be associated with an increased risk of asthma (OR = 1.81, 95% CI = 1.18–2.78).

KEYWORDS

adults, asthma, children, formaldehyde, indoor, meta-analysis

1 | INTRODUCTION

Asthma is a chronic inflammatory disorder arising from heterogenic gene-environment interactions, affecting 339 million people worldwide.^{1,2} With its well-known symptoms of wheezing dyspnea, chest tightness, coughing, and other prominent symptoms, asthma has a substantial impact on the quality of life.^{3,4} A study conducted in Asia revealed that short-term exposure to air pollution is associated with increased risk of asthma in children rather than in individuals aged 15–64 years, suggesting that differences may exist in asthma

between children and adults, and children may be more susceptible to environmental pollution due to their incompletely developed respiratory system.^{5,6}

Accumulating evidence shows that indoor environmental pollutants, such as respirable particles, pollen, mold, and house dust-mites, play important roles in the progression and occurrence of asthma, as 90% of human activities take place in the indoor environment (eg, home, classroom, and workplace).^{7–11} In the indoor environment, the concentration of formaldehyde (FA), which is a ubiquitous human carcinogenic (group 1) mainly released from tobacco smoke and

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furniture, has increased exceedingly due to lack of proper ventilation and high humidity.^{12,13} Recently, some studies were conducted to assess the relationship between FA exposure and asthma in children and adults,¹⁴⁻¹⁸ but the results were controversial, and these studies have specific limitations including potential selection bias and small sample size.

To the best of our knowledge, only a meta-analysis conducted by McGwin¹⁹ has ever reviewed and reported the pooled odds ratio (OR) with 95% confidence interval (CI) for the association between FA exposure and risk of asthma in children. After that, epidemiological evidence for such association has continued to accumulate over the nearly past decade,^{13,16,20-25} and no meta-analysis has ever systematically reviewed the effect of indoor FA exposure on asthma among adults. Therefore, in order to identify whether indoor exposure of FA is associated with asthma and provide scientific reference for preventive and mitigation actions, we conducted the present meta-analysis to systematically evaluate the association between indoor FA exposure and risk of asthma in both children and adults.

2 | METHODS

2.1 | Information sources and search terms

This study was conducted according to the Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines.²⁶ All papers selected in this study were searched from PubMed, Embase, and Web of Science. The search terms used for each database included: ("wheeze", "cough", "asthma", "asthmatic", "bronchial asthma") and ("formaldehyde", "formalin", and "methanol"). Studies were searched for the period up to January 1, 2020. Furthermore, the reference lists of relevant articles were conducted forward and backward citation chasing.

2.2 | Eligibility criteria

Studies were included if they meet the following criteria: (a) indoor exposure to FA; (b) the health effect was asthma; (c) the population-based studies designed as a cohort, cross-sectional, or case-control studies. The exclusion criteria were as follows: (a) no FA exposure; (b) asthma was not studied; (c) no OR, risk ratio (RR) or hazard ratio (HR), or primary data used to calculate these effects were reported; (d) clinical trials, reviews, letters to the editor, commentaries, and papers not written in English were excluded.

2.3 | Data extraction

Results of searching papers were merged in Endnote (version X8.2) to remove all duplicates. After screening titles and abstracts of the retrieved papers, data were extracted independently by two authors (LL.Y. and BW) according to a pre-designed form. Information on

Practical Implications

- The findings of this meta-analysis provide a scientific basis for targeted prevention and mitigation actions in indoor environments.
- Children are more prone to be adversely affected by FA than adults.

the name of the first author, publication year, region, study design, number of participants and cases, sex, asthma definition, monitoring point, the concentration of FA, adjusted confounders, and related effect estimates (ORs, RRs, and HRs, with 95% confidence interval CIs) were collected.

2.4 | Quality assessment

The Newcastle-Ottawa scale (NOS) were used to assess information bias, selection bias, and confounders in case-control and cohort studies²⁷ (Table 1). We assessed each cross-sectional study by the critical tools of the Joanna Briggs Institute (JBI),²⁸ which was used to evaluate the bias through a checklist of eight questions, as specified in Table 2. For the eight questions, if the proportion of answer "yes" $\geq 50\%$, then the relevant paper was considered as with low risk of bias; if the answer "no" $\geq 50\%$, then the risk of bias is high; and if the answer "unclear" $\geq 50\%$, then the risk of bias is uncertain. Two authors (YL and WB) assessed the methodological quality of each study independently. Any discrepancy between the two investigators was solved by discussing it with the senior investigator (WH.C).

2.5 | Statistical analysis

We sorted the FA concentrations reported in the selected articles and categorized exposure variables into low ($FA \leq 22.5 \mu\text{g}/\text{m}^3$) and high exposure ($FA > 22.5 \mu\text{g}/\text{m}^3$) according to the median concentration of FA. Besides, if the median or the mean FA concentration was not reported, the midpoint of the upper and lower boundaries was considered as its dose. An OR with 95% CI was used to estimate the asthma risk corresponding to each $10 \mu\text{g}/\text{m}^3$ increase in FA concentration. We used a random-effect model to calculate the combined ORs and its 95% CI for both children and adults.

Cochran's Q test and I^2 statistics were used to evaluate the heterogeneity, where $P < .10$ and $I^2 > 50\%$ were considered to substantial heterogeneity.²⁹ Subgroup analysis was conducted to explore the source of heterogeneity in both children and adults. Sensitivity analyses were performed by leave-one-out analyses from each study.³⁰ Evidence of publication bias or reporting bias was assessed by using the funnel plot along with the Eggers' test.³¹ All analyses were conducted using Stata version 15.1 (Stata Corp). All tests were

TABLE 1 Summarized characteristics of included studies in meta-analysis

Study	Region	Study type	Age group	Monitoring point	FA ($\mu\text{g}/\text{m}^3$)	FA level	Sample size	Asthma definition	OR (95% CI)	Confounding factor	QA ^a
Hulin et al (2010) ²¹	The Occident	Case-control	Child	Home	15.7 (4.8-63.1)	Low	51	Affirmative response to one of the following questions: Has your child ever had asthma?	10.70 (1.69-67.61)	Age, sex, family history of allergy, ETS, allergic rhinitis, location, season	8
	The Occident	Case-control	Child	Home	20.6 (3.7-60.3)	Low	63	Has your child had wheezing in the chest in the past 12 mo? Whether children have already taken medications against asthma crises?	1.07 (1.01-1.13)		
Fransman et al (2003) ³²	The Occident	Case-control	Adult	Factory	80 $\pm$ 3.0	High	527	Using the ECRHS definition: woken by shortness of breath in the last 12 mo; asthma attack in the last 12 mo; current asthma medication;	1.50 (0.9-2.8)	Age, sex, ethnicity	8
Zhai et al (2013) ²⁰	The east	Case-control	Child	Home	41-101	High	82	American Thoracic Society Adult/ Children Questionnaire (ATS-DLD-78)	4.25 (2.06-8.75)	Age, sex	7
	The east	Case-control	Adult	Home	41-101	High	186		2.60 (1.77-3.83)		
Araki et al (2012) ²²	The east	Cohort study	Adult	Home	32.6	High	161	Received medical treatment for bronchial asthma during the preceding two years	1.15 (0.26-5.08)	Age, sex, ETS, renovation history, wall-to-wall carpeting, dampness index, hay-fever.	7
Smedje et al (2001) ³⁶	The Occident	Cohort study	Child	School	8	Low	1258	Positive response to the following two questions: whether the pupil had ever had asthma?	1.20 (0.80-1.70)	Age, sex, ETS, history of allergy	8
								Was the diagnosis made by a physician?			
Matsunaga et al (2008) ²⁵	The east	Cohort study	Adult	Home	18-27	Low	299	Received medical treatment for asthma during the previous 12 mo	0.8 (0.23-2.84)	Age, sex, parity, family history of allergy, ETS, pets, mite, family income, education, season	8
	The east	Cohort study	Adult	Home	28-46	High	301		0.72 (0.19-2.77)		
	The east	Cohort study	Adult	Home	≥ 47	High	100		2.15 (0.41-11.28)		
Huang et al (2016) ¹⁶	The east	Cross-sectional	Child	Home	21.5 $\pm$ 13.0	Low	454	Ever diagnosed with asthma by a doctor	1.21 (0.60-2.45)	Age, sex, location, family history of allergy, ETS, season	/

(Continues)

TABLE 1 (Continued)

Study	Region	Study type	Age group	Monitoring point	FA ($\mu\text{g}/\text{m}^3$)	FA level	Sample size	Asthma definition	OR (95% CI)	Confounding factor	QA ^a
Kim et al (2011) ²³	The east	Cross-sectional	Child	School	27.7 ± 8.3	High	1915	Either having current asthma medication or having had an asthma attack during the last 12 mo	1.04 (0.78-1.4)	Age, sex, self-reported allergy, ETS	/
Zhao et al (2008) ³³	The east	Cross-sectional	Child	School	2.3 ± 1.1	Low	1993	Doctor-diagnosed asthma; Current asthma attacks; Current asthma medication	0.81 (0.49-1.33)	Age, sex, family history of allergy, ETS	/
Mi et al (2006) ³⁴	The east	Cross-sectional	Child	School	9.4 ± 6.9	Low	1414	Using the ECRHS definition: having either current asthma medication; having had an asthma attack during the previous 12 mo	1.30 (0.72-2.32)	Age, gender, ETS	/
Yon et al (2019) ¹³	The east	Cross-sectional	Child	School	27.17 ± 7.72	High	427	Defined by the International Study of Asthma and Allergies in Childhood questionnaire	1.02 (0.96-1.09)	Age, sex, ETS, family history of allergy	/
Smedje et al (1997) ³⁵	The Occident	Cross-sectional	Child	School	<5	Low	627	Ever had asthma diagnosed by a physician	1.10 (1.01-1.20)	Age, sex	/
Bentayeb et al (2015) ²⁴	The Occident	Cross-sectional	Adult	Home	7.21	Low	600	Positive response to the following two questions: has your doctor or another healthcare provider ever told you that you had asthma?	0.44 (0.12-1.56)	Age, sex, location, BMI, education background, smoking, season, ventilation	/

Note: OR (95%CI) presented here were estimates corresponding to a $10 \mu\text{g}/\text{m}^3$ increase of FA.

^aCase-control and cohort studies were assessed using the NOS, and cross-sectional studies were evaluated using another method without scores—JBI.

TABLE 2 Risk bias for cross-sectional studies

	Huang et al 2016	Kim et al 2011	Zhao et al 2008	Mi et al 2006	Yon et al 2019	Smedje et al 1997	Bentayeb et al 2015
Were the study subjects and the setting described in detail?	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Were the outcomes measured in a valid and reliable way?	Yes	Yes	Yes	No	Yes	No	Yes
Were the criteria for inclusion in the sample clearly defined?	Yes	Yes	Yes	Yes	Yes	No	Yes
Were strategies to deal with confounding factors stated?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Were objective, standard criteria used for measurement of the condition?	Yes	Yes	No	Yes	Yes	Unclear	Yes
Were confounding factors identified?	Yes	Yes	No	Yes	Yes	No	Yes
Was the exposure measured in a valid and reliable way?	Unclear	Yes	Yes	Unclear	Yes	Yes	Yes
Was appropriate statistical analysis used?	Yes	Unclear	Yes	Yes	Yes	Yes	Yes
Overall appraisal	Include	Include	Include	Include	Include	Include	Include

two-sided, and the significance level was defined as $P < .05$, except where otherwise specified.

3 | RESULTS

3.1 | Literature search

We initially retrieved a total of 1831 studies, of which 675 duplicates were removed. After screening through titles and abstracts, 1129 citations were excluded on account of review, non-human study, and irrelevant studies. The remaining 27 articles were reviewed in full text, and 14 of which were excluded as they did not report sufficient data or meet the inclusion criteria. Finally, 13 papers were included in our present meta-analysis. The above-mentioned process of study identification and selection is shown in Figure 1.

3.2 | Study description

As summarized in Table 1, of the total 13 included papers,^{13,16,20-25,32-36} 7 were cross-sectional studies, 3 were case-control studies and 3 were cohort studies. Among these studies, 8 were conducted in children, 4 in adults, and 1 was conducted in both children and adults. As to the study region, 8 studies were carried out in East and 5 in Occident. As to monitoring sites, 6 studies were conducted at home, 6 at school, and 1 study was conducted at a factory. Notably, of the 13 articles, one case-control study displayed two reports of effect estimates for different populations (children and adults), another case-control study displayed two reports of effect estimates for two case-control children populations, and a cohort study displayed three reports of effect estimates for different FA exposure levels. Therefore, there were 17 reports of effect estimates for the total 13 selected papers, where a total of 10 458 individuals were involved in with the sample size ranged from 51 to 1993. The good quality was detected for the case-control and cohort studies (mean quality score was 8; Table 1), and low risk of bias was detected for the cross-sectional studies (proportion of answer "yes" $\geq 50\%$, Table 2 and Figure 2).

3.3 | Overall analysis

Results from the random-effect model showed that, each 10 $\mu\text{g}/\text{m}^3$ increase in FA exposure was associated with a 10% (pooled OR = 1.10, 95% CI = 1.00-1.21, $I^2 = 63.10\%$, $P = .004$) significant increase in risk of asthma in children, while a 33% (pooled OR = 1.33, 95% CI = 0.79-2.22, $I^2 = 51.30\%$, $P = .055$) insignificant increase in risk of asthma in adults (Table 3). High heterogeneity ($I^2 > 50$) was found at home both in children and adults, and significant heterogeneity also found in child case-control studies (all conducted at home). We also observed significant heterogeneity in the different regions in both children and adults except for adults in East.

3.4 | Subgroup analyses

Subgroup analyses were conducted by FA concentration ($>/\leq 22.5 \mu\text{g}/\text{m}^3$), study design (cross-sectional/case-control/cohort studies), study region (the Occident/the East), and monitoring point (home/school or factory) in both children and adults, as presented in Table 3. In children, we found a significant association between FA exposure and increased risk of asthma only in subgroups of studies with cross-sectional design (OR = 1.05, 95% CI = 1.00-1.10, $I^2 = 0.0\%$, $P = .626$), studies conducted at school (OR = 1.05, 95% CI = 1.00-1.10,

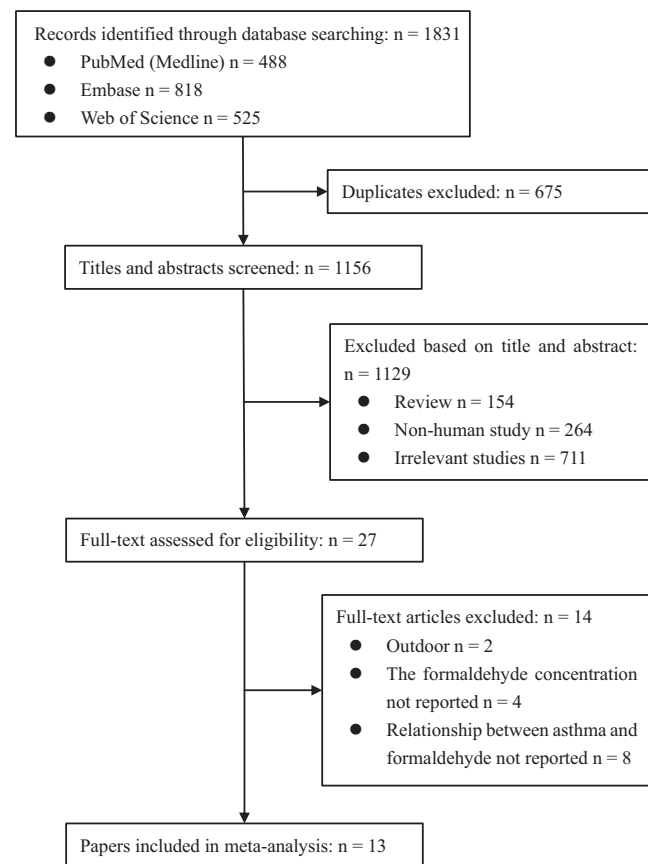


FIGURE 1 Flow diagram of literature search and study selection

$I^2 = 0.00\%$, $P = .577$), and studies with low ($\leq 22.5 \mu\text{g}/\text{m}^3$) FA exposure level (OR = 1.09, 95% CI = 1.00-1.19, $I^2 = 27.5\%$, $P = .219$). Notably, the positive association of FA exposure with asthma was stronger in subgroups of studies with high ($>22.5 \mu\text{g}/\text{m}^3$) FA exposure level (OR = 1.39, 95% CI = 0.87-2.19, $I^2 = 86.5\%$, $P = .001$); however, this association was lack of statistical significance probably due to the poor statistical power result from the insufficient articles (only 3 reports).

In adults, a significant association of FA exposure with increased risk of asthma was only found in subgroups of studies with case-control design (OR = 2.06, 95% CI = 1.21-3.51, $I^2 = 59.7\%$, $P = .115$), and studies with high ($>22.5 \mu\text{g}/\text{m}^3$) FA exposure level (OR = 1.81, 95% CI = 1.18-2.78, $I^2 = 27.9\%$, $P = .235$). Moreover, the association of FA exposure with asthma was more evident in subgroups of studies with high rather than low ($\leq 22.5 \mu\text{g}/\text{m}^3$) FA exposure level.

3.5 | Sensitivity analysis and publication bias

Sensitivity analysis was conducted by excluding each study in turn, which did not alter the combined OR. There was no significant publication bias observed using Egger's ($P = .094$ and 0.051 , respectively).

4 | DISCUSSION

To our knowledge, this paper is the first meta-analysis to study the relationship between indoor FA exposure and asthma in both adults and children. A significant association between indoor FA exposure and increased risk of asthma in children was found, and children exposed to a higher level of FA were associated with a higher risk of asthma. Besides, we also found a significantly increased risk of asthma in adults with high concentrations of FA exposure.

There may be several reasons for the difference in results between adults and children. Firstly, children spend more time indoor than adults, leading to more indoor FA exposure and may result in more evident asthmatic effect.¹ Secondly, children are generally

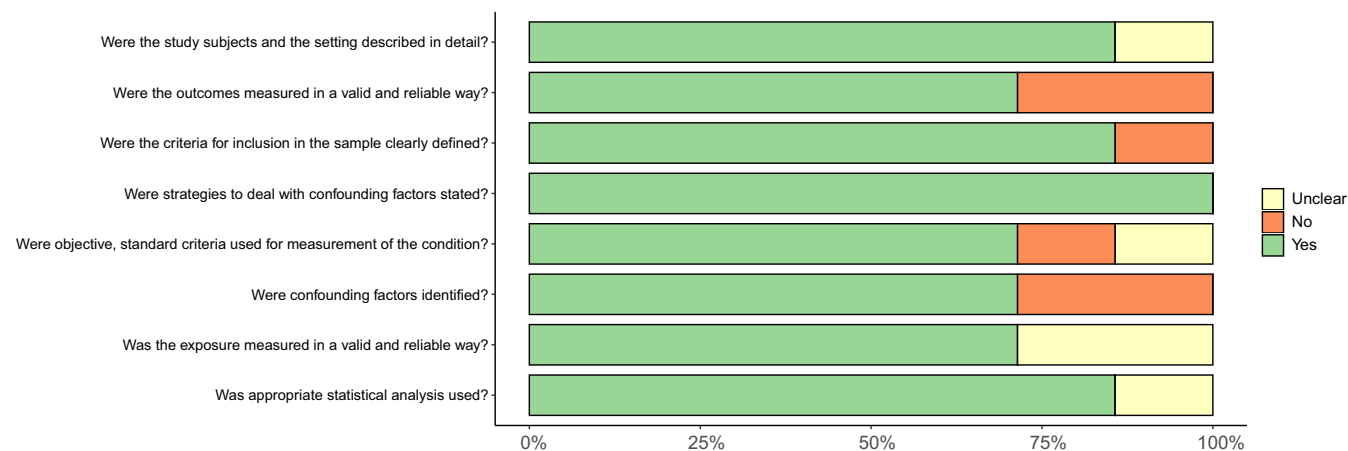


FIGURE 2 The proportion of studies with low, high, or unclear risk of bias

TABLE 3 Results of subgroup analysis in children and adults

Children	No. reports	Odds ratio	95% confidence interval	I^2	P for heterogeneity
All	10	1.10	1.00-1.21	63.10%	.004
FA concentration classification					
Low exposure ($\leq 22.5 \mu\text{g}/\text{m}^3$)	7	1.09	1.00-1.19	27.50%	.219
High exposure ($> 22.5 \mu\text{g}/\text{m}^3$)	3	1.39	0.87-2.19	86.50%	.001
Study design					
Case-control studies	3	2.96	0.81-10.80	89.90%	<.001
Cross-sectional studies	6	1.05	1.00-1.10	0.00%	.626
Cohort studies	1	1.20	0.82-1.75	/	/
Region					
The Occident	4	1.10	0.99-1.22	53.90%	.089
The East	6	1.20	0.91-1.59	69.70%	.006
Monitoring point					
Home	4	2.14	0.94-4.88	85.00%	<0.001
School	6	1.05	1.00-1.10	0.00%	0.577
Adults	No. reports	Odds ratio	95% confidence interval	I^2	P for heterogeneity
All	7	1.33	0.79-2.22	51.30%	.055
FA concentration classification					
Low exposure ($\leq 22.5 \mu\text{g}/\text{m}^3$)	2	0.60	0.24-1.46	0.00%	.514
High exposure ($> 22.5 \mu\text{g}/\text{m}^3$)	5	1.81	1.18-2.78	27.90%	.235
Study design					
Case-control studies	2	2.06	1.21-3.51	59.70%	.115
Cross-sectional studies	1	0.44	0.12-1.59	/	/
Cohort studies	4	1.01	0.50-2.04	0.00%	.752
Region					
The Occident	2	0.93	0.29-3.01	66.00%	.087
The East	5	1.54	0.83-2.88	38.80%	.162
Monitoring point					
Home	6	1.19	0.59-2.41	57.80%	.037
Factory	1	1.50	0.85-2.65	/	/

more susceptible to air pollution than adults, on account of a faster breathing rate of children, and their respiratory volume is 50% more than adults.³⁷ Thirdly, due to the immature physiological system, children may be more susceptible to the adverse effect of FA.³⁷ According to the World Health Organization indoor air guidelines,³⁸ the limit of indoor FA concentration is $0.1 \text{ mg}/\text{m}^3$; however, our results show that children exposed to low concentrations of FA ($\leq 22.5 \mu\text{g}/\text{m}^3$) were significantly associated with increased risk of asthma. Therefore, measures should be taken to reduce the indoor FA concentration and protect children from asthma, such as choosing environmentally friendly decorative materials, adequate ventilation, and installing cleaning equipment. In adults, the number of articles in the low concentration group was only two, and the meta-result may be resultantly biased.

Significant heterogeneity was found in this meta-analysis and should be considered with caution. First of all, in children, significant

heterogeneity was observed at high-concentration exposures. Concentration, the main factor involved in the health effect of FA, may also have caused the heterogeneity. Therefore, for understanding the relationship of FA with asthma conclusively, research considering exposure of high FA concentration exposure in children is warranted. Next, the associations varied by the monitoring point. We observed that there is no heterogeneity at school, while heterogeneity is high for both adults and children at home. The reason may be due to different choices of locations for monitoring FA concentration in the home (eg, living room, bedroom), and the number of participants at school was significantly higher than at home. There is only one selected article in the indoor workplace,³² and more research is needed. Zhai et al²⁰ and Smedje et al³⁵ considered only sex and age as confounding factors. Most of the confounding factors, such as family history of allergy and environmental tobacco smoke (ETS), were included in other selected studies. Differences between

confounding factors might also lead to heterogeneity. In our research, significant heterogeneity between regions may be due to different races³⁹; many other confounding factors, such as season,⁴⁰ dampness,⁴¹ and mites,⁴² should be considered in future studies.

Asthma is divided into allergic and non-allergic. Allergic asthma is triggered by allergens and related to an allergic reaction, while non-allergic asthma is not related to allergens. The role of FA sensitization in the pathogenesis of asthma has been widely addressed.⁴³ FA is a small molecule with the ability to conjugate with large protein molecules in serum (eg, albumin). Such combination provoking the formation of IgE antibodies, then leading to degranulation of mast cells with allergic response.⁴⁴ Moreover, FA could increase T-helper cell type 2(Th2) polarization through an adjuvant effect, thus leading to cytokine mediators (eg, IL4, IL5, and IL13) release, epithelial mucous cell metaplasia, and airway recruitment of eosinophils.⁴⁴ Besides, a recent study found that FA exposure could induce IL-17, NF- κ B elevation, and MAPK activation, suggesting that not all asthmatic patients have purely type 2 inflammation, but also non-type 2 inflammation.⁴⁵

Our study also has some limitations. First, due to the small number of papers selected for children and adults, some groups have very few papers, and there may be bias. Second, inconsistent definition of asthma in each article. Third, most of the selected papers were cross-sectional studies, which cannot provide evidence in causality. Fourth, a dose-response relationship could not be conducted, as most selected articles had only one dose level of FA. Well-designed prospective epidemiologic studies with large sample size and longer follow-up duration are warranted to identify the association and causality between indoor FA exposure and asthma in both children and adults.

5 | CONCLUSION

This meta-analysis suggests that exposure to indoor FA was associated with increased risk of asthma in children, and such positive association was persistently significant under an indoor FA concentration ($\leq 22.5 \mu\text{g}/\text{m}^3$) much lower than the WHO recommendation ($0.1 \text{ mg}/\text{m}^3$). There may be an association between high concentrations of FA exposure and asthma in adults. Thus, our research findings provide a scientific reference for the adverse effects of FA on human health and make a case for immediate, targeted actions of prevention, and mitigation in indoor environments.

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CONFLICT OF INTEREST

None of the authors declares any conflict of interest.

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